Chapter 8: Closed-form Planck Boundaries

Before we can construct a world from five Planck boundaries, using the binomial constructor and the hyperbolic partition equation, we must have precise closed-form definitions for those boundaries.

We've already seen that the constants of Nature repeatedly point toward the five Planck boundaries: t_p, l_p, q_p, T_p, and m_p. These are the boundaries of coherent atomic logic, appearing again and again within the ratios and transformations that make physics work. Yet despite their foundational role, these boundaries are not known with full precision.

That blurring of precision occurs because each Planck boundary is reconstructed from relationships among multiple constants of Nature. Those limiting relationships yield values for the Planck boundaries that agree to roughly seven digits, then diverge. This divergence is a signal. It occurs at the same order as the inversion boundary, $\boxtimes \approx 1 \times 10^{-7}$, which is why the seventh digit marks the transition between the leading Planck-bounded layer and the internal inversion-boundary layer.

If we are to build a unified system from Planck boundaries, we cannot rely on incomplete inputs. If the Planck boundaries are truly structural, then their ultimate definitions cannot depend on which reconstruction pathway we choose. They should be defined, not just partially known. And that means they should be given precise closed-form definitions.

That became the mission: to hunt for closed forms that could serve as candidate definitions of the Planck boundaries themselves. Not approximations. Exact geometric actions that return each Planck boundary's value. If the Planck boundaries are real—if they are the structural anchors of Nature—then there must exist a geometry that pins them in place.

So the search began.

The search was algorithmic. I converted each Planck boundary into a dimensionless target, separated its significand from its signed decimal exponent, and then searched for simple closed forms capable of binding those two pieces together. I sought simple algebraic-geometric expressions that characterized something modular or structurally meaningful: expressions composed from geometric elements such as e, π, and functions like $sin(x)$, $cos(x)$, $sinh(x)$, or $cosh(x)$, provided they reproduced the reported digits of the Planck boundaries and could be made coherent as a set.

I used WolframAlpha to identify closed-form expressions that returned the experimentally observed values, and then systematically tested whether those expressions—when treated as exact—could reproduce the constants of Nature through the binomial constructor and the hyperbolic partition equation.

Of course, precision alone wasn't enough. I was searching for five expressions that worked together—five closed forms that shared the same kind of internal logic, respected the same symmetry framework, and hinted at a single geometric ancestry.

The search was therefore narrowed by two key constraints: simplicity, where expressions with cleaner algebraic structure were favored; and hyperbolic character, where expressions built from elements that appeared naturally in hyperbolic geometry were favored. I then used the closed forms I found as exact inputs, building the constants of Nature from them to test whether they revealed a coherent pattern among the constants.

To make this search concrete, each Planck boundary is normalized by its base unit and separated into three numerical features: its partially known digit string, its base-ten scale, and the signed order of magnitude carried by that scale.

Let B_k denote the k^{th} Planck boundary:

$$\begin{aligned}
B_0 &= t_p \\
B_1 &= l_p \\
B_2 &= q_p \\
B_3 &= T_p \\
B_4 &= m_p
\end{aligned}$$

Each boundary is made dimensionless by measuring it relative to its base unit:

$$\begin{aligned}
u_0 &= \text{second} \\
u_1 &= \text{meter} \\
u_2 &= \text{coulomb} \\
u_3 &= \text{kelvin} \\
u_4 &= \text{kilogram}
\end{aligned}$$

The numerical value of each normalized Planck boundary B_k/u_k can then be written in scientific notation as

$$\frac{B_k}{u_k} = \phi_k \times 10^{n_k}.$$

Here, $\phi_k \in [1,10)$ is the significand: the partially known digit string. The factor 10^{n_k} gives the base-ten scale. The integer n_k is the signed order of magnitude carried by that scale.

The known normalized values of the five Planck boundaries are:

$$\begin{aligned}
&m_p/\mathrm{kg} = 2.176434(24) \times 10^{-8} \\
&T_p/\mathrm{K} = 1.416784(16) \times 10^{32} \\
&q_p/\mathrm{C} = 1.8755459 \times 10^{-18} \\
&l_p/\mathrm{m} = 1.616255(18) \times 10^{-35} \\
&t_p/\mathrm{s} = 5.391247(60) \times 10^{-44}
\end{aligned}$$

The corresponding digit strings and signed integer exponents are:

$$\begin{aligned}
\phi_4 &= 2.176434(24) & n_4 &= -8 \\
\phi_3 &= 1.416784(16) & n_3 &= +32 \\
\phi_2 &= 1.8755459 & n_2 &= -18 \\
\phi_1 &= 1.616255(18) & n_1 &= -35 \\
\phi_0 &= 5.391247(60) & n_0 &= -44
\end{aligned}$$

To probe this structure, we ask what signed dimensionless geometric factor G_k must be paired with the exponential generator e^{ϕ_k} in order to reproduce the signed integer n_k associated with each Planck boundary.

$$G_k\, e^{\phi_k} = n_k \qquad \text{equivalently} \qquad G_k = \frac{n_k}{e^{\phi_k}}.$$

Here G_k is the signed geometric factor, ϕ_k is the dimensionless significand, and n_k is the signed decimal exponent selected by the boundary.

This rewrites each partially known Planck value as a balance between a continuous geometric expression and a discrete integer target.

To unveil each Planck boundary's geometric factor G_k, I plugged in the respective ϕ_k values and searched for simple algebraic-geometric expressions that characterize the resulting signed numbers.

$$G_4 = -8/e^{2.176434(24)} \quad \text{Planck mass}$$
$$G_3 = +32/e^{1.416784(16)} \quad \text{Planck temperature}$$
$$G_2 = -18/e^{1.8755459} \quad \text{Planck charge}$$
$$G_1 = -35/e^{1.616255(18)} \quad \text{Planck length}$$
$$G_0 = -44/e^{5.391247(60)} \quad \text{Planck time}$$

This time, a matched set appeared. The search returned five simple, logically related hyperbolic expressions—one for each Planck boundary. Each expression reproduces the corresponding CODATA value within its reported uncertainty, and because each is written in closed form, each candidate value can be evaluated to arbitrary precision.

For compactness, the closed forms below are shown after canceling common negative signs where they occur.

candidate closed-forms

$$5\left(\frac{4\pi}{2}\right)\left(\cos\left(\frac{7}{5}\right)\right)^2 = \frac{8}{e^{\Phi_4}} \qquad \text{mass}$$

$$2\left(\frac{5}{\sqrt{7}}\right)^2\left(\cos\left(\frac{5}{2}i\right)\right)^2\left(\cos\left(\frac{7}{5}\right)\right)^2 = \frac{32}{e^{\Phi_3}} \qquad \text{temperature}$$

$$\left(\frac{5}{\sqrt{7}}\right)\left(\frac{3^{-1/3}}{W_{We}}\right) = \frac{18}{e^{\Phi_2}} \qquad \text{charge}$$

$$\left(\sinh\left(\sinh\left(\frac{1}{7}\right)\right)\right)^{-1} = \frac{35}{e^{\Phi_1}} \qquad \text{length}$$

$$\left(\frac{4\pi}{4}\right)\left(\sinh\left(\frac{1}{4}\right)\right)^2 = \frac{!5}{e^{\Phi_0}} \qquad \text{time}$$

Here $cos(x)$ is the cosine function, $sinh(x)$ is the hyperbolic sine function, π = Archimedes' constant, i = the imaginary unit, W_{We} = the Weierstrass constant, and $!5 = 44$. More generally, $!n$ denotes the derangement (sub-factorial) function:

$$!n = n!\sum_{k=0}^{n}\frac{(-1)^k}{k!}.$$

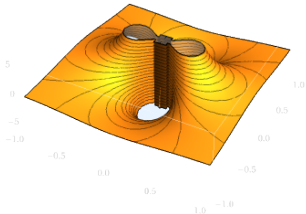
Digit Comparison	$\sigma = 1.00$, standard deviation
$m_p = 2.1764268381757 \times 10^{-8}$ kg $m_p = 2.176434(24) \times 10^{-8}$ kg	$\sigma = 0.33$, CODATA 2022
$T_p = 1.4167869859079 \times 10^{32}$ K $T_p = 1.416784(16) \times 10^{32}$ K	$\sigma = 0.13$, CODATA 2022
$q_p = 1.8755459671396 \times 10^{-18}$ C $q_p = 1.8755459 \times 10^{-18}$ C	$\sigma = 0.00$, CODATA 2022
$l_p = 1.6162591817564 \times 10^{-35}$ m $l_p = 1.616255(18) \times 10^{-35}$ m	$\sigma = 0.22$, CODATA 2022
$t_p = 5.3912583683231 \times 10^{-44}$ s $t_p = 5.391247(60) \times 10^{-44}$ s	$\sigma = 0.18$, CODATA 2022

In each pair, the first value is the candidate closed-form value and the second value is the reported CODATA value.

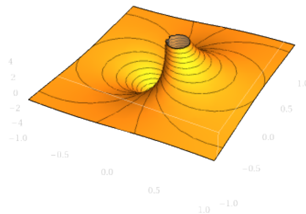
The five candidate boundary expressions therefore share a common action-balance form. Each boundary is selected where a continuous geometric expression balances the integer scale carried by its decimal exponent, and each reproduces the CODATA value for a Planck boundary within the reported 1σ uncertainty.

To visualize the behavior of these definitions, we extend their trigonometric or hyperbolic arguments into the complex plane.[60] The resulting surface plots allow us to diagnose the analytic character of the candidate boundary definitions and compare their geometric behavior.

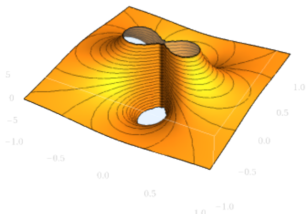
These surface plots expose strikingly common structure. Each surface is organized by lemniscate-like contours, with one branch ascending and another descending through the central region. The Planck length surface is shaped by a split lemniscate connection along its central spine, while the Planck time, charge, temperature, and mass surfaces are composed of paired ascending and descending lemniscate connections. The Planck mass surface is oriented orthogonally to the other four, and the Planck charge surface has the simplest self-intersection.



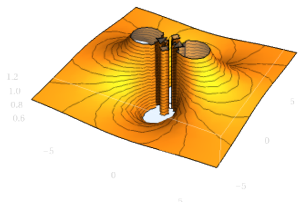
Planck time



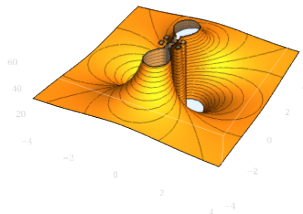
Planck length



Planck charge



Planck temperature



Planck mass

Together, these expressions exhibit a shared coherence. They all take the same action-balance form, $G_k\, e^{\Phi_k} = n_k$, and draw from the same geometric ingredients—$cos(x)$, $sinh(x)$, π, e, i, and the Weierstrass constant W_{We}. These ingredients fit naturally into a single architectural scheme. This internal consistency was precisely what I had been searching for.

$$i^{-i} = \frac{s^4\, {W_{We}}^4}{8} \qquad e^{\pi} = \frac{s^8\, {W_{We}}^8}{8^2} \qquad s = \frac{\Gamma^2\left(\frac{1}{4}\right)}{\sqrt{2\pi}}$$

$$e^{\pi} = 4\, L^8\, {W_{We}}^8 \qquad s = 2L$$

The Weierstrass constant—W_{We}—is interwoven with π, e, i, through the arc length of the unit lemniscate—s—the lemniscate constant—L, and the gamma function $\Gamma(x)$. These relations place these candidate Planck boundary definitions within a unified web of classical geometric constants.

Of course, this does not mean these are the true definitions of the Planck boundaries. Not yet. At this stage, they are still hypotheses—tools in the toolkit. But they are simple and exact tools, from which we can now attempt to compose the constants of Nature.

To earn their place in the theory, these candidate definitions must do more than match the five Planck values. They must help reconstruct the constants of Nature as one organized system. Only then can the boundaries justify themselves structurally.

With definitions in hand, I began using the Planck boundaries to deconstruct every constant of Nature through the logic of the binomial constructor and the hyperbolic partition equation.

I had tried many possible sets of Planck-boundary definitions by this point, finding few clues to follow. But with this coherent set of definitions, a pattern began to emerge—one that, over the next couple of years, expanded to encompass all 288 constants of Nature.

With each new constant decoded the pattern became clearer—and more striking. The constants of Nature began to appear to be woven into a single structure. And the geometric components responsible for maintaining each boundary within that structure traced to a common origin.

Not only did this structure cohere; its geometric elements appeared to trace back to a very special geometry: the simplest possible arena constructible as a 3-manifold. The hunt for Planck boundaries—the boundaries built into the rules of physics—led us to the simplest geometric stage: a manifold capable of carrying structure, transformation, and persistence.

The next chapter examines the three constants of Nature that assemble the Dirac equation, and asks whether the geometry of the simplest constructible 3-manifold reveals itself in their structure.